

ARBATROSS

Installation & User Guide

Overview

ARBATROSS is a web application built with **Next.js** and **Three.js**. This guide explains how to install, run, and troubleshoot the project.

1. Prerequisites

Install Node.js

Before installing ARBATROSS, make sure Node.js is available on your device.

Windows / macOS

Download the latest LTS version from nodejs.org.

Linux (Ubuntu / Debian)

```
sudo apt install nodejs npm
```

Android (Termux)

```
pkg install nodejs
```

Verify Installation

```
node -v
```

```
npm -v
```

If both commands return version numbers, the installation was successful.

2. Project Installation

Step 1 — Open the Project Directory

```
cd Arbatross
```

Step 2 — Install Dependencies

```
npm install
```

This command downloads and installs all required packages listed in `package.json`.

3. Running ARBATROSS

Development Mode

```
npm run dev
```

Open the application in your browser:

```
http://localhost:3000
```

Benefits: Automatic reload on file changes Faster development workflow Real-time updates

Production Mode

```
npm run build
```

```
npm run start
```

Use this mode to test the optimized production build before deployment.

4. Important Project Files

src/app/globals.css

Global styles, theme colors, typography, and layout settings.

src/components/

Contains Dashboard, MET display, telemetry panels, and UI components.

src/components/OrbitalGlobe.jsx

Controls the 3D Earth visualization and orbital rendering.

public/

Stores images, Earth textures, icons, and 3D models (.glb).

5. Troubleshooting

Problem: npm command not found

Solution: Reinstall Node.js and restart your terminal.

Problem: Permission Denied (EACCES)

Solution: Avoid using sudo unless required and ensure the project directory belongs to your user account.

Problem: Black Screen or Missing 3D Scene

Solution: Verify WebGL support Open Developer Tools (F12) Check browser console errors Ensure assets inside the public folder are loading correctly